

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I Ankala Lakshmi (192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Lakshmi*

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.
5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

A retail company operating physical stores and an online platform generates data from sales transactions, inventory systems, customer accounts, websites, mobile applications, and loyalty programs. The main difficulty is that this data comes from different sources and at different speeds. If inventory information is not updated quickly, the company may show products as available online even when they are out of stock. Similarly, without proper customer analysis, marketing campaigns may be generic instead of personalized.

The architecture should have five major layers:

Data Sources → Data Ingestion → Storage → Processing & Analytics → Business Applications

Layer	Technologies	Purpose
Data Sources	POS, website, mobile app, CRM, inventory systems	Generate sales and customer data
Ingestion	Apache Kafka, APIs	Collect batch and real-time data
Storage	HDFS, Amazon S3, Azure Data Lake	Store large-scale structured/unstructured data
Processing	Apache Spark, Spark Streaming	Clean and process data
Analytics	ML, Spark MLlib, Python	Forecast demand and customer behavior
Visualization	Power BI/Tableau	Display inventory and business insights

Sales transactions from physical stores and online purchases can be sent to Apache Kafka as real-time events. Kafka acts as a buffer between data-generating applications and the processing system. Historical data can be stored in a cloud-based data lake such as Amazon S3 or Azure Data Lake.

Apache Spark can then process both historical and streaming data. Data cleaning can remove duplicate transactions, incorrect records, and missing values. The processed information can be stored in an analytical database or warehouse for reporting.

Analytics Models: For demand forecasting, time-series models such as ARIMA, Prophet, or machine-learning models such as Random Forest or Gradient Boosting can be used. The model can consider previous sales, seasonality, promotions, holidays, and location. For personalized marketing, collaborative filtering and content-based recommendation models can identify products that are likely to interest each customer. For example, if a customer regularly buys sports shoes and fitness accessories, the system can recommend new sports products rather than sending unrelated advertisements.

2. Social Media Fraud Detection Problem

A social media platform receives billions of posts, comments, likes, shares, follows, and login events every day. The major challenge is that fraudulent accounts and misinformation can spread very quickly. A traditional system that processes data only once a day would be too slow.

The solution should therefore use a real-time stream-processing architecture.

Proposed Architecture

User Activity → Kafka → Stream Processing → Feature Extraction → ML Models → Risk Score → Action

Component	Suggested Technology	Role
Event collection	Apache Kafka	Collect user activities
Stream processing	Apache Flink / Spark Streaming	Process events continuously
Storage	Data Lake / HDFS	Store historical information
ML	Python, Scikit-learn, TensorFlow	Detect suspicious behavior
Graph analysis	Neo4j / Graph processing	Identify connected fake accounts
Search	Elasticsearch	Fast searching and investigation
Dashboard	Power BI/Tableau	Monitor suspicious activity

Whenever a user creates a post, shares content, or performs another activity, the event can be sent to Kafka. A framework such as Apache Flink can process these events with very low latency.

The system can calculate features such as:

- Number of posts per minute
- Sudden increase in followers
- Repeated sharing of identical content
- Unusual login locations
- Account age
- Ratio of followers to following
- Interaction patterns
- Connection with previously blocked accounts

Machine learning can then assign a fraud or misinformation probability.

For example, an account created two hours ago that suddenly follows thousands of users and posts hundreds of identical messages could receive a high-risk score.

Role of Machine Learning

Different models can be used depending on the problem:

Problem	Possible Technique
Fake account detection	Random Forest, XGBoost
Abnormal behavior	Anomaly detection
Text misinformation	NLP models
Coordinated accounts	Graph analytics
Spam detection	Classification models

The system should not automatically delete every suspicious post because false positives are possible. Instead, high-risk cases can be blocked temporarily or sent for human review.

3. Government Data Platform Problem

A national government data platform has to combine information from different departments such as census, taxation, economics, transportation, healthcare, and geographic information systems. The major challenge is that each department may use different formats, databases, naming conventions, and security policies.

For example, one department may identify a district using a numerical code while another may use its name. Without standardization, combining these datasets can produce incorrect results.

Therefore, the solution should focus not only on technology but also on data governance, interoperability, security, and accountability.

Proposed Data Management Strategy

The architecture can be organized as:

Department Systems → Secure Data Ingestion → Data Lake → Standardization → Shared Data Platform
→ Controlled Access

Requirement	Strategy
Data integration	APIs, ETL/ELT pipelines
Storage	Government cloud/data lake
Standardization	Common schemas and data formats
Metadata	Central data catalog

Security	Encryption + IAM
Privacy	Anonymization/masking
Access	Role-based access control
Quality	Validation and auditing
Interoperability	Standard APIs and common data models

A data lake can store large volumes of census, economic, geospatial, and departmental data. ETL/ELT pipelines can transform the data into standardized formats before it is shared. A centralized metadata catalog should describe where the data came from, who owns it, when it was updated, and what it can be used for.

Data Governance

A governance committee can establish common standards for:

- Data ownership
- Data quality
- Data classification
- Data retention
- Data sharing
- Privacy
- Security
- Access permissions

Sensitive citizen information should not be freely available to every department. Role-Based Access Control (RBAC) should ensure that users can access only the information required for their responsibilities. Personally identifiable information should be protected using techniques such as encryption, masking, pseudonymization, and anonymization where appropriate.

Interoperability

Government departments should communicate through standardized APIs and common data models rather than creating isolated systems.

4. Banking Fraud Detection Problem

A bank processing millions of transactions every minute cannot depend on batch processing because fraudulent transactions need to be identified before or immediately after they occur. At the same time, analyzing only current transactions is not enough. The system must also understand the customer's historical behavior.

Therefore, the best solution is a hybrid batch and real-time big data pipeline.

Proposed Pipeline

Transactions → Kafka → Stream Processing → Feature Engineering → Fraud Model → Risk Score → Decision

Historical data can follow a separate path:

Historical Transactions → Data Lake → Spark → Model Training → Updated ML Model

Stage	Technology	Purpose
Transaction collection	Kafka	Receive transactions continuously
Stream processing	Flink/Spark Streaming	Real-time analysis
Historical storage	HDFS/S3	Store large datasets
Processing	Apache Spark	Large-scale processing
Database	SQL/NoSQL	Store customer and transaction information
ML	Random Forest/XGBoost	Fraud classification
Monitoring	Dashboard/Alerts	Track suspicious transactions

When a transaction occurs, information such as transaction amount, location, merchant, device, time, and customer history can be passed to Kafka.

A streaming engine such as Apache Flink can process the transaction and calculate relevant features within milliseconds or seconds.

For example, consider a customer who normally spends ₹2,000–₹5,000 in Chennai. Suddenly, a ₹1,00,000 transaction occurs from another country. The system can identify this as unusual when compared with the customer's historical behavior.

Algorithms

Several techniques can be combined:

- Random Forest / XGBoost — classification
- Logistic Regression — baseline fraud probability
- Isolation Forest — anomaly detection

- Clustering — identify unusual transaction groups
- Rule-based detection — immediate checks for known fraud patterns

A hybrid approach is better than depending on one algorithm.

Why Streaming Is Important

Batch processing might identify fraud several hours later. By then, the money may already be transferred.

Streaming allows the bank to:

1. Receive the transaction.
2. Analyze it immediately.
3. Calculate the risk score.
4. Approve, reject, or hold the transaction.

5. Agriculture Smart Farming Problem

Modern agriculture can generate large amounts of data from soil sensors, weather stations, drones, satellites, GPS devices, irrigation systems, and farm equipment. The challenge is to combine these distributed data sources and convert them into useful recommendations for farmers.

The objective is not simply to collect data. The real objective is to answer practical questions such as:

- When should irrigation be performed?
- Which area needs more fertilizer?
- Is the crop under stress?
- What yield can be expected?
- Is there a possibility of pest or disease attack?

Proposed Architecture

Sensors/Drones/Satellites → IoT Gateway → Kafka → Cloud Data Lake → Spark → Analytics/ML → Farmer Dashboard

Data Source	Information Generated
Soil sensors	Moisture, pH, nutrients
Weather sensors	Temperature, humidity, rainfall
Drones	Crop images and plant health

Satellites	Large-scale field imagery
GPS	Vehicle and field location
Farm equipment	Fuel and machine activity

Sensor data can be continuously transmitted through an IoT gateway. Apache Kafka can handle the incoming stream of events. Historical information can be stored in a cloud data lake such as Amazon S3 or Azure Data Lake.

For large-scale processing, Apache Spark can combine soil, weather, crop, and historical yield data.

Analytics and Machine Learning

Different analytics techniques can solve different agricultural problems.

Requirement	Analytics Technique
Yield prediction	Regression / Random Forest
Disease detection	Image classification
Irrigation planning	Predictive analytics
Weather impact	Time-series analysis
Crop grouping	Clustering
Soil analysis	Classification/regression
Resource optimization	Optimization algorithms

For example, if soil moisture falls below a particular level and weather data shows no expected rainfall, the system can recommend irrigation for that specific field area. Drone images can also be analyzed using computer vision to identify unhealthy crop regions. Instead of applying fertilizer equally across the entire farm, farmers can focus resources on areas that actually require treatment.

Benefits

The system can help farmers:

- Reduce water consumption
- Reduce unnecessary fertilizer usage
- Detect crop diseases earlier
- Predict crop yield
- Improve resource allocation
- Increase productivity

- Reduce operational costs

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